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Interconnect structure and electronic device including the same

Abstract

Provided are an interconnect structure and an electronic device including the same. The interconnect structure may include a conductive wiring having a certain pattern, a dielectric layer on side surfaces of the conductive wiring, a capping layer on the conductive wiring, and a graphene layer on the dielectric layer. The graphene layer may include a graphene material. A ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene material is 1 or less.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0082332, filed on Jun. 24, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

(2) The present disclosure relates to an interconnect structure and an electronic device including the same.

2. Description of the Related Art

(3) In recent years, the size of semiconductor devices has been gradually decreased for the high integration of semiconductor devices. To this end, the line width of a conductive wiring including a metal or a metal alloy in an interconnect structure may need to be reduced. When the line width of the conductive wiring is reduced, the current density in the conductive wiring increases, and as a result, the resistance of the conductive wiring increases. This increase in the resistance of the conductive wiring may cause electromigration of the atoms of the metal or the metal alloy constituting the conductive wiring, resulting in defects in the conductive wiring. Therefore, a capping layer capable of improving the resistance of the conductive wiring is needed.

(4) In accordance with the miniaturization of semiconductor devices, the parasitic capacitance of a dielectric layer may be an important factor in improving the performance of wirings. The dielectric layer may include a low-dielectric constant (low-k) material and fine pores to adjust the dielectric constant. Moisture may permeate the fine pores in the dielectric layer from the outside, resulting in an increase in the dielectric constant, and thus, the effect of using the low-dielectric constant (low-k) material may be impaired.

SUMMARY

(5) An aspect of example embodiments is to provide an interconnect structure in which a change in a dielectric constant of a dielectric layer is limited and/or prevented and the dielectric constant of the dielectric layer is maintained due to the hydrophobic surface characteristics of a graphene layer.

(6) Another aspect is to provide an interconnect structure in which a capping layer is provided to reduce the resistance of a conductive wiring.

(7) Still another aspect is to provide an electronic device including the interconnect structure.

(8) Additional aspects will be set forth in part in the description that follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments of the disclosure.

(9) According to an embodiment, an interconnect structure may include a conductive wiring having a certain pattern, a dielectric layer on side surfaces of the conductive wiring, a capping layer on the conductive wiring, and a graphene layer on the dielectric layer. The graphene layer may include a graphene material, and a ratio of carbons having $sp^{3.5}$ bonds to carbons having sp^2 bonds in the graphene material may be 1 or less.

(10) In some embodiments, the graphene layer may have a contact angle of about 80° to about 110° .

(11) In some embodiments, the capping layer and the graphene layer may include different materials.

(12) In some embodiments, the capping layer and the graphene layer may include a same material.

(13) In some embodiments, the graphene layer may include about 1 atomic % to about 20 atomic % hydrogen.

(14) In some embodiments, the graphene layer may further include one or more of F, Cl, Br, and N, and a content of the one or more of F, Cl, Br, and N may be in a range of about 1 atomic % to about 20 atomic %.

(15) In some embodiments, an intensity ratio of a D-band peak to a G-band peak in a Raman

spectrum of the graphene layer may be 3.5 or less.

(16) In some embodiments, an intensity ratio of a 2D-band peak to a G-band peak in a Raman spectrum of the graphene layer may be 0.2 or greater.

(17) In some embodiments, an average grain size of the graphene material may be about 0.5 nm to about 100 nm.

(18) In some embodiments, a thickness of the graphene layer may be about 1 nm to about 20 nm.

(19) In some embodiments, a density of the graphene layer may be about 1.6 g/cc to about 2.1 g/cc.

(20) In some embodiments, the conductive wiring may include one or more of Cu, Ru, Rh, Ir, Mo, W, Pd, Pt, Co, Ta, and Ti, or an alloy thereof, or a subcombination thereof, or a combination thereof.

(21) In some embodiments, the interconnect structure may further include a barrier layer between the dielectric layer and the conductive wiring.

(22) In some embodiments, the interconnect structure may further include a liner layer between the conductive wiring and the barrier layer.

(23) According to an embodiment, an electronic device may include a substrate, and an interconnect structure on the substrate. The interconnect structure may include a conductive wiring having a certain pattern, a dielectric layer on side surfaces of the conductive wiring, a capping layer on the conductive wiring, and a graphene layer on the dielectric layer. The graphene layer may include a graphene material. A ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene material may be 1 or less.

(24) In some embodiments, the graphene layer may have a contact angle of about 80° to about 110°.

(25) In some embodiments, the capping layer and the graphene layer may include different materials.

(26) In some embodiments, the capping layer and the graphene layer may include a same material.

(27) In some embodiments, the graphene layer may include about 1 atomic % to about 20 atomic % hydrogen.

(28) In some embodiments, the graphene layer may further include one or more of F, Cl, Br, and N, and a content of the one or more of F, Cl, Br, and N may be in a range of about 1 atomic % to about 20 atomic %.

(29) In some embodiments, an intensity ratio of a D-band peak to a G-band peak in a Raman spectrum of the graphene layer may be 3.5 or less.

(30) In some embodiments, an intensity ratio of a 2D-band peak to a G-band peak in a Raman spectrum of the graphene layer may be 0.2 or greater.

(31) According to an embodiment, an interconnect structure may include a dielectric layer including a trench, a graphene layer on a first surface of the dielectric layer, a conductive wiring in the trench, and a capping layer on the conductive wiring. The trench may be recessed in a first direction from the first surface of the dielectric layer to a second surface of the dielectric layer. The trench may have a pattern. The graphene layer may define an opening over the trench. A ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene layer may be 1 or less.

(32) In some embodiments, the capping layer and the graphene layer may include different materials, and the capping layer may be in the opening defined by the graphene layer.

(33) In some embodiments, the dielectric layer may include at least one of Al₂O₃, AlN, ZrO₂, HfO_x (0<x<4), SiO₂, SiCO, SiCN, SiON, SiCOH, AlSiO, or boron nitride (BN). The graphene layer may be configured to limit impurities from entering the dielectric layer through the first surface of the dielectric layer.

(34) In some embodiments, the graphene layer may further include one or more of F, Cl, Br, and N. A content of the one or more of F, Cl, Br, and N may be in a range of about 1 atomic % to about 20 atomic %.

(35) In some embodiments, a barrier layer may be in the trench between the dielectric layer and the conductive wiring.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view of an interconnect structure according to an embodiment;
- (3) FIG. 2 is a cross-sectional view of the interconnect structure illustrated in FIG. 1, taken along line A-A';
- (4) FIGS. 3A to 3D are schematic diagrams illustrating a method of manufacturing a graphene layer, according to an embodiment;
- (5) FIG. 3E is a diagram illustrating an X-ray photoelectron spectroscopic (XPS) analysis result with respect to a graphene layer, according to an embodiment;
- (6) FIG. 4 is a diagram illustrating a Raman spectroscopic spectrum analysis result with respect to a graphene layer, according to an embodiment;
- (7) FIG. 5 is a transmission electron microscopic (TEM) image of an interconnect structure according to an embodiment;
- (8) FIG. 6A is a picture of a dielectric layer with moisture thereon, according to a comparative example;
- (9) FIG. 6B is a picture of a graphene layer with moisture thereon, according to an embodiment;
- (10) FIG. 6C is a picture of a graphene layer including fluorine with moisture thereon, according to another embodiment;
- (11) FIG. 7A is a cross-sectional view of an interconnect structure according to another embodiment; FIG. 7B is a cross-sectional view of an interconnect structure according to another embodiment;
- (12) FIG. 8 is a perspective view of an interconnect structure according to an embodiment;
- (13) FIG. 9 is a cross-sectional view of the interconnect structure illustrated in FIG. 8, taken along line B-B';
- (14) FIG. 10 is a TEM image of an interconnect structure according to an embodiment;
- (15) FIG. 11A is a cross-sectional view of an interconnect structure according to another embodiment;
- (16) FIG. 11B is a cross-sectional view of an interconnect structure according to another embodiment;
- (17) FIG. 12 is a schematic diagram of an electronic device according to an embodiment; and
- (18) FIGS. 13A and 13B are cross-sectional views of electronic devices including interconnect structures according to example embodiments.

DETAILED DESCRIPTION

(19) Reference will now be made in detail to embodiments, examples of which are illustrated in the accompanying drawings, wherein like reference numerals refer to like elements throughout. In this regard, the present embodiments may have different forms and should not be construed as being limited to the descriptions set forth herein. Accordingly, the embodiments are merely described below, by referring to the figures, to explain aspects. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. Expressions such as “at least one of,” when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, “at least one of A, B, and C,” “at least one of A, B, or C,” “one of A, B, C, or a combination thereof,” and “one of A, B, C, and a combination thereof,”

respectively, may be construed as covering any one of the following combinations: A; B; C; A and B; A and C; B and C; and A, B, and C.”

(20) Hereinafter, an interconnect structure according to an embodiment of the present disclosure, an electronic device including the interconnect structure, and a method of manufacturing the interconnect structure will be described in detail with reference to the accompanying drawings. The descriptions below are merely non-limiting examples, and the present disclosure is not limited thereto and is only defined by the scope of claims described below. In the drawings, like reference numerals denote like elements, and the size or thickness of each element may be exaggerated for clarity and convenience of description.

(21) As used herein, the term “combination” includes a mixture, an alloy, a reaction product, or a composite form with an element unless otherwise stated. Throughout the present specification, when an element “includes” an element, unless there is a particular description contrary thereto, the element may further include other elements, not excluding the other elements. An expression used in the singular encompasses the expression of the plural, unless otherwise indicated or it has a clearly different meaning in the context. The term “or” refers to “and/or” unless otherwise stated.

(22) As used herein, the terms “an embodiment”, “embodiments”, and the like indicate that elements described with regard to an embodiment are included in at least one embodiment described in this specification and may or may not present in other embodiments. In addition, it should be understood that the described elements may be combined in any suitable manner in various embodiments. When an element is referred to as being “on” another element, it may be directly on the other element or intervening elements may be present therebetween. Unless otherwise defined, technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this application belongs. All patents, patent applications, and other references cited herein are incorporated herein by reference in their entirety. However, in the event of any conflict or inconsistency between terms used herein and terms of the cited references, the terms used in this specification take precedence over the terms of the cited references.

(23) When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value includes a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical value. Moreover, when the words “generally” and “substantially” are used in connection with geometric shapes, it is intended that precision of the geometric shape is not required but that latitude for the shape is within the scope of the disclosure. Further, regardless of whether numerical values or shapes are modified as “about” or “substantially,” it will be understood that these values and shapes should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated numerical values or shapes.

(24) While particular embodiments are described, alternatives, modifications, variations, improvements, and substantial equivalents that are or may be presently unforeseen may arise to applicants or those of ordinary skill in the art. Accordingly, the appended claims as filed and as they may be amended are intended to embrace all such alternatives, modification, variations, improvements, and substantial equivalents.

(25) The term “graphene” used herein refers to a sheet structure of a single atomic layer formed by a carbon nanostructure that forms polycyclic aromatic molecules in which a plurality of carbon atoms are covalently bound and aligned in a planar shape; a network structure in which a plurality of carbon nanostructures as a small film having a plate shape are interconnected and aligned in a planar shape; or a combination thereof. The covalently bound carbon atoms may form 6-membered rings as repeating units, but may further include 5-membered rings and/or 7-membered rings. The graphene may be formed by stacking a plurality of layers including several sheet structures and/or network structures, and the average thickness may be about 100 nm or less, for example, about 10 nm or less, specifically, about 0.01 nm to about 10 nm.

(26) FIG. 1 is a perspective view of an interconnect structure **10** according to an embodiment. FIG. 2 is a cross-sectional view of the interconnect structure **10** illustrated in FIG. 1, taken along line A-A'.

(27) Referring to FIGS. 1 and 2, the interconnect structure **10** according to an embodiment may include a graphene layer **100** on a dielectric layer **200**, a conductive wiring **300**, and a capping layer **400**. The dielectric layer **200** may be on side surfaces of a conductive wiring **300**. The conductive wiring **300** may have a certain pattern. The capping layer **400** may be on the conductive wiring **300**.

(28) The dielectric layer may define a trench T that may be recessed from a first surface (e.g., upper surface) of the dielectric layer **200** to a second surface (e.g., bottom surface of trench T) of the dielectric layer **200**. The graphene layer **100** may be on the first surface of the dielectric layer **200** and may define an opening O over the trench O. The capping layer **400** may be in the opening O defined by the graphene layer **100**.

(29) The graphene layer **100** according to an embodiment may include a graphene material, and the ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene material may be 1 or less. The graphene layer **100** may be on the dielectric layer **200** to limit and/or block permeation of impurities, for example, moisture, etc., into the dielectric layer **200**. A technical feature of protecting the dielectric layer **200** by using the graphene layer **100** will be described below with reference to FIGS. 3A to 6C.

(30) The dielectric layer **200** may have a single-layer structure or a multi-layer structure in which different materials are stacked. The dielectric layer **200** according to an embodiment may include a dielectric material that is used in a general semiconductor manufacturing process. In accordance with the miniaturization of semiconductor devices, the parasitic capacitance of the dielectric layer **200** may be an important factor in improving the performance of wirings. Accordingly, the dielectric layer **200** may include a low-dielectric constant (low-k) material as an inter-metal dielectric (IMD). The dielectric layer **200** according to an embodiment may include pores for dielectric constant adjustment.

(31) For example, the dielectric layer **200** may include a dielectric material having a dielectric constant of about 3.6 or less, for example, about 3.3 or less, for example, about 3.0 or less. For example, the dielectric layer **200** may include at least one of Al₂O₃, AlN, ZrO₂, HfO_x (0<x<4), SiO₂, SiCO, SiCN, SiON, SiCOH, AlSiO, or boron nitride (BN). However, the present disclosure is not limited thereto, and various other dielectric materials may be used in the dielectric layer **200**. In addition, the dielectric layer **200** may include an organic dielectric material.

(32) The conductive wiring **300** may include a metal or a metal alloy having excellent electrical conductivity. The conductive wiring **300** may include one or more metals selected from Cu, Ru, Rh, Ir, Mo, W, Pd, Pt, Co, Ta, and Ti, or an alloy thereof, or a subcombination thereof, or a combination thereof. In accordance with the miniaturization of semiconductor devices, the conductive wiring **300** may have a nano-scale line width, but is not limited thereto. For example, the dielectric layer **200** may be on side surfaces or the bottom surface of the conductive wiring **300**, but not on the conductive wiring **300**. Throughout the present specification, regions in which the dielectric layer **200** may be provided, e.g., those on the side surfaces or the bottom surface of the conductive wiring **300**, are defined as regions excluding a region on the top surface of the conductive wiring **300** in which the capping layer **400** may be provided.

(33) The capping layer **400** may be on the conductive wiring **300** to reduce the electrical resistance of the conductive wiring **300**, thereby improving the electromigration resistance. The capping layer **400** may include a material different from that of the graphene layer **100**. For example, the capping layer **400** may include a conductive material such as cobalt or tungsten.

(34) For example, in the case where the capping layer **400** formed of cobalt (Co) is on the conductive wiring **300** composed of copper wires, the electrical resistance of the conductive wiring

300 may be reduced by about 4% or more, compared to the case where the capping layer **400** is not provided on the conductive wiring **300**.

(35) As described above, the dielectric layer **200** may include the fine pores to adjust the dielectric constant. Impurities, for example, moisture, etc., may permeate the fine pores included in the dielectric layer **200** from the outside. This results in an increase in the dielectric constant of the dielectric layer **200**, and thus the effect of using the low-dielectric constant (low-k) material may be impaired. Hereinafter, the graphene layer **100**, which may be on the dielectric layer **200** to limit and/or block permeation of impurities, for example, moisture, etc., into the dielectric layer **200**, will be described.

(36) FIGS. **3A** to **3D** are schematic diagrams illustrating a method of manufacturing a graphene layer, according to an embodiment. FIG. **3E** is a diagram illustrating an X-ray photoelectron spectroscopic (XPS) analysis result on a graphene layer, according to an embodiment. FIG. **4** is a diagram illustrating a Raman spectroscopic spectrum analysis result on a graphene layer, according to an embodiment. FIG. **5** is a transmission electron microscopic (TEM) image of an interconnect structure according to an embodiment. FIG. **6A** is a picture of a dielectric layer with moisture thereon, according to a comparative example. FIG. **6B** is a picture of a graphene layer with moisture thereon, according to an embodiment. FIG. **6C** is a picture of a graphene layer including fluorine with moisture thereon, according to another embodiment.

(37) The graphene layer **100** according to an embodiment may be grown on the surface of the dielectric layer **200** by atomic layer deposition (ALD), chemical vapor deposition (CVD), or plasma-enhanced CVD (PECVD). In the present embodiment, the graphene layer **100** may not be formed on the surface of the conductive wiring **300**, but may be formed only on the surface of the dielectric layer **200**. Hereinafter, a method of forming the graphene layer **100** on the dielectric layer **200** by performing PECVD described above will be described with reference to FIGS. **3A** to **3D**.

(38) Referring to FIG. **3A**, power for plasma generation is applied into a reaction chamber C, in which the dielectric layer **200** and the conductive wiring **300** are provided, and a reaction gas for growing the graphene layer **100** is injected into the reaction chamber.

(39) Then, power for plasma generation is applied from a plasma power source (not shown) into the reaction chamber C. Here, the power for plasma generation may be about 10 W to about 3000 W. However, the present disclosure is not limited thereto.

(40) For example, a radio frequency (RF) power source or a microwave (MW) power source may be used as the plasma power source.

(41) When the power for plasma generation is applied to the reaction chamber C from the plasma power source, an electric field may be induced in the reaction chamber C. In such a state in which the electric field is induced, the reaction gas for growing the graphene layer **100** is injected into the reaction chamber C.

(42) A mixed gas of a carbon source gas, an inert gas, and a hydrogen gas may be used as the reaction gas for growing the graphene layer **100**. The carbon source gas may be a gas which supplies carbon for growth of nanocrystalline graphene. For example, the carbon source gas may include at least one of a hydrocarbon gas and a vapor of a liquid precursor including carbon. The hydrocarbon gas may include, for example, at least one of methane gas, ethylene gas, and propylene gas, but is not limited thereto. In addition, the liquid precursor may include, for example, at least one of toluene, benzene, anisole, and ethanol, but is not limited thereto. The inert gas may include, for example, at least one of argon gas, nitrogen gas, helium gas, krypton gas, and xenon gas. FIG. **3A** illustrates a case where the carbon source gas is acetylene gas, and the inert gas is argon gas.

(43) The process temperature for growing the graphene layer **100** may be about 500° C. or less, which is less than a temperature used in a general chemical vapor deposition process. For example, the process temperature inside the reaction chamber may be about 200° C. to about 500° C.

(44) In addition, the process pressure for growing the graphene layer **100** may be about 0.01 Torr to

about 1 Torr. However, this is merely a non-limiting example, and other process pressures may be used.

(45) When the reaction gas in which the carbon source gas, the inert gas, and the hydrogen gas are mixed is introduced into the reaction chamber C, the reaction gas is converted into a plasma state by an electric field applied by plasma power.

(46) Referring to FIG. 3B, active carbon radicals C. are produced by plasma of the reaction gas in which the carbon source gas, the inert gas, and the hydrogen gas are mixed, and are adsorbed onto the surface of the dielectric layer **200**. Specifically, plasma of the inert gas in the reaction gas produces the active carbon radicals C. from the carbon source gas, the produced active carbon radicals C. are adsorbed onto the surfaces of the dielectric layer **200** and the conductive wiring **300** as illustrated in FIG. 3C, and thus the graphene layer **100** is grown within a short period of time.

(47) Referring to FIG. 3D, a portion of the graphene layer **100** on the conductive wiring **300** may be removed by performing an etching process, to form the graphene layer **100** only on the dielectric layer **200**.

(48) The graphene material included in the graphene layer **100** according to an embodiment may include an edge carbon (edge C) that exists at an edge site and a center carbon (center C) that exists at a center site. The edge carbon has a sp³ bonding structure, and the center carbon has a sp² bonding structure. The sp³ bonding structure is a three-dimensional bonding structure of a regular tetrahedron of carbons such as diamond, and the sp² bonding structure is a two-dimensional bonding structure of graphite. Because a functional group such as oxygen or nitrogen may be bound to the edge carbon, the edge carbon may have a higher reactivity to an etching solution compared to the center carbon. The ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene material may be 1 or less.

(49) The graphene layer **100**, in which the amount of carbons having sp² bonds is larger than the amount of carbons having sp³ bonds, may have a higher contact angle than that of the dielectric layer **200**. For example, the surface of the graphene layer **100**, having a ratio of carbons having sp³ bonds to carbons having sp² bonds of 1 or less, may have a high contact angle of about 80° to about 110°. However, the present disclosure is not limited thereto. The term 'contact angle' used herein refers to the water contact angle (WCA) of a surface. The water contact angle refers to the angle of contact between a drop of water and an air-liquid-solid interface.

(50) According to an experimental result on the contact angle, as shown in FIG. 6A, the contact angle of the dielectric layer **200** was measured to be approximately 34.8°. In this regard, as shown in FIG. 6B, the contact angle of the graphene layer **100** was measured to be approximately 90.6°.

(51) The fact that the graphene layer **100** in which the amount of carbons having sp² bonds is larger than the amount of carbons having sp³ bonds has a higher contact angle than that of the dielectric layer **200** means that the graphene layer **100** has a stable surface having lower surface energy than that of the dielectric layer **200**. Accordingly, the graphene layer **100** on the dielectric layer **200** may protect the dielectric layer **200** from permeation of impurities from the outside, for example, moisture, etc., into the dielectric layer **200**. Accordingly, the dielectric constant of the dielectric layer **200** may be limited and/or prevented from being increased, and thus degradation of the electrical characteristics of the interconnect structure may be limited and/or prevented.

(52) The graphene layer **100** may further include atoms having high electronegativity to further reduce the surface energy. For example, F, Cl, Br, N, P, or O atoms may be additionally included in the graphene layer **100** through surface treatment of the graphene layer **100**. In this case, the graphene layer **100** may have a higher contact angle than when the above-described atoms are not added. For example, as shown in FIG. 6C, the contact angle of the graphene layer **100** including F atoms was measured to be approximately 102.1°.

(53) The ratio of carbons having a sp² bonding structure to total carbons may be confirmed by performing XPS analysis. The XPS analysis was performed by using Quantum 2000 (Physical Electronics) (acceleration voltage: 0.5 keV to 15 keV, 300 W, energy resolution: about 1.0 eV,

sputter rate: 0.1 nm/min). The result of Raman analysis, as illustrated in FIG. 3E, shows that a graphene material having an sp² bonding structure predominates in the graphene layer **100** according to an embodiment. From this, it may be seen that the ratio of carbons having sp³ bonds to carbons having sp² bonds in the graphene layer **100** is 1 or less.

(54) The average grain size of the graphene material of the graphene layer **100** according to an embodiment may be 0.5 nm or more and 100 nm or less. Here, the average grain size refers to D50, and indicates the grain size when the graphene material in which grains with various sizes are distributed is accumulated up to 50% by volume. Here, the average grain size may be an average particle diameter when the graphene material is spherical, and may be the length of the long axis when the graphene material is not spherical (e.g., oval or rectangular).

(55) In addition, the thickness of the graphene layer **100** according to an embodiment may be about 1 nm to about 20 nm. For example, the thickness of the graphene layer **100** may be about 1 nm to about 10 nm, about 1 nm to about 5 nm, or about 1 nm to about 3 nm. Here, the graphene layer **100** may have a density of about 1.6 g/cc to about 2.1 g/cc. The graphene material has a stable structure when the thickness and density of the graphene layer **100** are in the respective ranges described above.

(56) Referring to FIG. 4, the graphene layer **100** according to an embodiment exhibits peaks at about 1340 cm⁻¹ to about 1350 cm⁻¹, about 1580 cm⁻¹, and about 2700 cm⁻¹ in the Raman spectrum. These peaks provide information of a thickness, a crystallinity, and a charge doping status of the graphene material. The peak observed at about 1580 cm⁻¹ is a “G-band” peak, which is generated in a vibration band corresponding to stretching of a carbon-carbon bond, and the energy of the “G-band” is determined according to the density of an excess of charges doped in the graphene material. The peak observed at about 1340 cm⁻¹ to about 1350 cm⁻¹ is a “D-band” peak, which appears when a sp² crystal structure has defects and which is mainly observed when many defects are in a sample or around edges of the sample. The peak observed at about 2700 cm⁻¹ is a “2D-band” peak which is useful in evaluating the thickness of the graphene material.

(57) The graphene layer **100** may have an intensity ratio of the D-band peak to the G-band peak of about 3.5 or less in the Raman spectrum.

(58) The graphene layer **100** may have an intensity ratio of the 2D-band peak to the G-band peak of about 0.2 or greater in the Raman spectrum.

(59) The graphene layer **100** may include hydrogen in an amount of about 1 atomic % to about 20 atomic %. The amounts of carbon and hydrogen may be confirmed by, for example, XPS analysis and Rutherford backscattering spectroscopic (RBS) component analysis. As described above, the graphene layer **100** may further include atoms having high electronegativity, for example, F, Cl, Br, N, P, or O atoms, to further reduce the surface energy. The concentration of atoms added to the graphene layer **100** may be about 1 atomic % to about 20 atomic %, but is not limited thereto.

(60) FIG. 7A is a cross-sectional view of the interconnect structure **10** according to another embodiment. FIG. 7B is a cross-sectional view of the interconnect structure **10** according to another embodiment.

(61) Referring to FIG. 7A, the interconnect structure **10** according to an embodiment may further include a barrier layer **500** between the dielectric layer **200** and the conductive wiring **300**. The barrier layer **500** according to an embodiment may be formed by performing a deposition process including CVD, PECVD, physical vapor deposition (PVD), ALD, sputtering, and/or plating, which are used in a general semiconductor manufacturing process. For example, the barrier layer **500** may include a metal, an alloy of a metal, a metal nitride, or the like. For example, the barrier layer **500** may include a conductive material such as Ti, Ta, Ru, W, Ir, RuTa, IrTa, TiN, TaN, RuN, RuTa_N, IrTa_N, WN, or graphene. However, the present disclosure is not limited thereto. The barrier layer **500** may have a single-layer structure or a multi-layer structure in which different materials are stacked. The thickness of the barrier layer **500** may be changed according to a material used as well

as the deposition process. For example, the thickness of the barrier layer **500** may be about 1 nm to about 40 nm.

(62) Referring to FIG. 7B, the interconnect structure **10** according to an embodiment may further include a liner layer **600** between the conductive wiring **300** and the barrier layer **500**. The liner layer **600** according to an embodiment may be formed by performing a deposition process including CVD, PECVD, PVD, ALD, sputtering, and plating, which are used in a general semiconductor manufacturing process. For example, the liner layer **600** may include a metal, an alloy of a metal, a metal nitride, or the like. For example, the liner layer **600** may include Ta, Ti, Co, TaN, Ti(Si)N, or W. However, the present disclosure is not limited thereto. The liner layer **600** may have a single-layer structure or a multi-layer structure in which different materials are stacked. The thickness of the liner layer **600** may be changed according to a material used as well as the deposition process. For example, the thickness of the liner layer **600** may be about 1 nm to about 40 nm. The liner layer **600** may include a different material than the barrier layer **500**.

(63) FIG. 8 is a perspective view of an interconnect structure **10'** according to an embodiment. FIG. 9 is a cross-sectional view of the interconnect structure **10'** illustrated in FIG. 8, taken along line B-B'. FIG. 10 is a TEM image of an interconnect structure according to an embodiment. FIG. 11A is a cross-sectional view of the interconnect structure **10'** according to another embodiment. FIG. 11B is a cross-sectional view of the interconnect structure **10'** according to another embodiment.

(64) Referring to FIGS. 8 and 9, the interconnect structure **10'** according to an embodiment may include the graphene layer **100** on the dielectric layer **200**, the dielectric layer **200** on side surfaces of the conductive wiring **300**, the conductive wiring **300** having a certain pattern, and the capping layer **400** on the conductive wiring **300**. The other components than the capping layer **400** are substantially the same as those illustrated in FIGS. 1 and 2, and thus descriptions thereof will be omitted for convenience of description.

(65) The capping layer **400** according to an embodiment may include the same material as that of the graphene layer **100**. For example, the capping layer **400** may include a graphene material in the same manner as the graphene layer **100**. In this case, the capping layer **400** may be integrally formed with the graphene layer **100**, on the conductive wiring **300**.

(66) Referring to FIG. 11A, the interconnect structure **10'** according to an embodiment may further include the barrier layer **500** between the dielectric layer **200** and the conductive wiring **300**. Referring to FIG. 11B, the interconnect structure **10'** according to an embodiment may further include the liner layer **600** between the conductive wiring **300** and the barrier layer **500**. The other components than the capping layer **400** are substantially the same as those illustrated in FIGS. 7A and 7B, and thus descriptions thereof will be omitted for convenience of description.

(67) FIG. 12 is a schematic diagram of an electronic device **1** according to an embodiment.

(68) Referring to FIG. 12, the electronic device **1** according to an embodiment may include a substrate G extending along one plane, and one or more interconnect structures **11** to **13** on the substrate G. The substrate G according to an embodiment may include at least one of a group IV semiconductor material, a semiconductor compound, an insulating material, and a metal. For example, the substrate G may include a group IV semiconductor material such as Si, Ge, or Sn. As another example, the substrate G may include at least one of Si, Ge, C, Zn, Cd, Al, Ga, In, B, C, N, P, S, Se, As, Sb, Te, Ta, Ru, Rh, Ir, Co, Ta, Ti, W, Pt, Au, Ni, and Fe. For example, the substrate G may include Si, Ge, SiC, SiGe, SiGeC, a Ge alloy, GaAs, InAs, InP, or the like. The substrate G may include a single layer or a plurality of layers in which different materials are stacked.

(69) For example, the substrate G may include a silicon-on-insulator (SOI) substrate or a silicon germanium-on-insulator (SGOI) substrate. For example, the substrate G may further include N and F as a SiCOH-based composition, and may include pores to lower the dielectric constant. Meanwhile, the substrate G may further include a dopant.

(70) The substrate G may include at least one semiconductor device (not shown). For example, the semiconductor device may include at least one of a transistor, a capacitor, a diode, and a resistor.

For example, the semiconductor device may be a transistor. However, the present disclosure is not limited thereto, and any semiconductor device that may be used in the art may be used.

(71) FIGS. **13A** and **13B** are cross-sectional views of electronic devices including interconnect structures according to example embodiments.

(72) Referring to FIG. **13A**, an electronic device **1300a** may include a device portion **210** having a semiconductor region **215**, and an element layer **205** on the semiconductor region **215**. The device portion **210** and semiconductor region **215** may be part of a circuit (e.g., logic circuit, memory circuit, image sensor circuit). The dielectric layer **200** may cover the device portion **210** and surround the element layer **205**. The dielectric layer **200** may define a hole or trench that exposes the semiconductor region **215** and element layer **205**. The conductive wiring **300** may be in the hole or trench and may be electrically connected to the element layer **205**.

(73) A barrier layer **500** may be in the hole or trench of the dielectric layer **200** between the conductive wiring **300** and the dielectric layer **200**. The barrier layer **500** may further extend between the conductive wiring **300** and the element layer **205**. A liner layer **600** may be in the hole or trench of the dielectric layer **200** and may be between the conductive wiring **300** and the barrier layer **500**. In some embodiments, the barrier layer **500** and/or liner layer **600** may be omitted. A graphene layer **100** may be formed on the dielectric layer **200** and define an opening over the hole or trench in which the conductive wiring **300** is disposed. A capping layer **400** may be formed on the conductive wiring **300** and may be positioned in the opening defined by the graphene layer **100**.

(74) In FIG. **13A**, the capping layer **400** may be a different material than the graphene layer **100**. Alternatively, as shown in FIG. **13B**, an electronic device **1300b** may be the same as the electronic device **1300a** in FIG. **13A**, except the capping layer **400** of the electronic device **1300b** may include the same material as a graphene layer **100**.

(75) An aspect of embodiments of inventive concepts provides an interconnect structure, in which a graphene layer has hydrophobic surface characteristics is on a dielectric layer and is capable of limiting and/or preventing impurities from permeating the dielectric layer so as to limit and/or prevent an increase in the dielectric constant of the dielectric layer.

(76) Another aspect of embodiments of inventive concepts provides an interconnect structure including a capping layer capable of reducing the resistance of a conductive wiring, so as to limit and/or prevent deterioration of electrical characteristics.

(77) Still another aspect provides is an electronic device including the interconnect structure described above.

(78) It should be understood that embodiments described herein should be considered in a descriptive sense only and not for purposes of limitation. Descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments. While one or more embodiments have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of inventive concepts as defined by the following claims.

Claims

1. An interconnect structure comprising: a conductive wiring having a certain pattern; a dielectric layer on side surfaces of the conductive wiring; a capping layer on the conductive wiring; and a graphene layer on the dielectric layer, wherein the graphene layer comprises a graphene material, and a ratio of carbons having sp^{sup.3} bonds to carbons having sp^{sup.2} bonds in the graphene material is 1 or less, wherein the capping layer and the graphene layer comprise different materials, wherein the graphene layer is on an upper surface of the dielectric layer and defines an opening over an upper surface of the conductive wiring, and wherein the capping layer is on the upper surface of the conductive wiring and in the opening defined by the graphene layer, such that the

graphene layer surrounds sidewalls of the capping layer.

2. The interconnect structure of claim 1, wherein the graphene layer has a contact angle of about 80° to about 110° .

3. The interconnect structure of claim 1, wherein a lower surface of the graphene layer, the upper surface of the conductive wiring, and a lower surface of the capping layer are coplanar.

4. The interconnect structure of claim 1, wherein the graphene layer comprises about 1 atomic % to about 20 atomic % hydrogen.

5. The interconnect structure of claim 1, wherein the graphene layer further comprises one or more of F, Cl, Br, and N, and a content of the one or more of F, Cl, Br, and N is in a range of about 1 atomic % to about 20 atomic %.

6. The interconnect structure of claim 1, wherein an intensity ratio of a D-band peak to a G-band peak in a Raman spectrum of the graphene layer is 3.5 or less.

7. The interconnect structure of claim 1, wherein an intensity ratio of a 2D-band peak to a G-band peak in a Raman spectrum of the graphene layer is 0.2 or greater.

8. The interconnect structure of claim 1, wherein an average grain size of the graphene material is about 0.5 nm to about 100 nm.

9. The interconnect structure of claim 1, wherein a thickness of the graphene layer is about 1 nm to about 20 nm.

10. The interconnect structure of claim 1, wherein a density of the graphene layer is about 1.6 g/cc to about 2.1 g/cc.

11. The interconnect structure of claim 1, wherein the conductive wiring comprises one or more of Cu, Ru, Rh, Ir, Mo, W, Pd, Pt, Co, Ta, and Ti, or an alloy thereof, or a subcombination thereof, or a combination thereof.

12. The interconnect structure of claim 1, further comprising: a barrier layer between the dielectric layer and the conductive wiring.

13. The interconnect structure of claim 12, further comprising: a liner layer between the conductive wiring and the barrier layer.

14. The interconnect structure of claim 1, wherein the upper surface of the dielectric layer and the upper surface of the conductive wiring are coplanar.

15. An electronic device comprising: a substrate; and an interconnect structure on the substrate, wherein the interconnect structure comprises a conductive wiring having a certain pattern, a dielectric layer on side surfaces of the conductive wiring, a capping layer on the conductive wiring, and a graphene layer on the dielectric layer, wherein the graphene layer comprises a graphene material, a ratio of carbons having sp^3 bonds to carbons having sp^2 bonds in the graphene material is 1 or less wherein the capping layer and the graphene layer comprise different materials, wherein the graphene layer is on an upper surface of the dielectric layer and defines an opening over an upper surface of the conductive wiring, and wherein the capping layer is on the upper surface of the conductive wiring and in the opening defined by the graphene layer, such that the graphene layer surrounds sidewalls of the capping layer.

16. The electronic device of claim 15, wherein the graphene layer has a contact angle of about 80° to about 110° .

17. The electronic device of claim 15, wherein a lower surface of the graphene layer, the upper surface of the conductive wiring, and a lower surface of the capping layer are coplanar.

18. The electronic device of claim 15, wherein the graphene layer comprises about 1 atomic % to about 20 atomic % hydrogen.

19. The electronic device of claim 15, wherein the graphene layer further comprises one or more of F, Cl, Br, and N, and a content of the one or more of F, Cl, Br, and N is in a range of about 1 atomic % to about 20 atomic %.

20. The electronic device of claim 15, wherein an intensity ratio of a D-band peak to a G-band peak in a Raman spectrum of the graphene layer is 3.5 or less.

21. The electronic device of claim 15, wherein an intensity ratio of a 2D-band peak to a G-band peak in a Raman spectrum of the graphene layer is 0.2 or greater.

22. The electronic device of claim 15, wherein the upper surface of the dielectric layer and the upper surface of the conductive wiring are coplanar.
